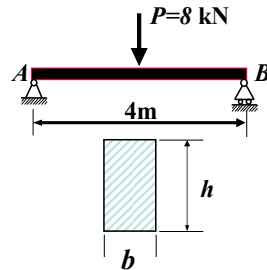


Lecture 18 – Bending – problem solving tutorial 2

1. A simple support beam as shown in the following figure, allowable normal stress is $[\sigma]=12\text{MPa}$, allowable shear stress is $[\tau]=1\text{MPa}$. Design height and width of cross section are h and b , respectively ($b/h=1.5$).



Solution:

1) check the normal stress

$$\sigma = \frac{M}{W} \leq [\sigma] \quad M = \frac{PL}{4} = 8\text{ kN}\cdot\text{m} \quad W = \frac{bh^2}{6} = \frac{1.5h^3}{6} = \frac{h^3}{4}$$

$$h \geq \sqrt[3]{\frac{4M}{[\sigma]}} = 138\text{ mm}$$

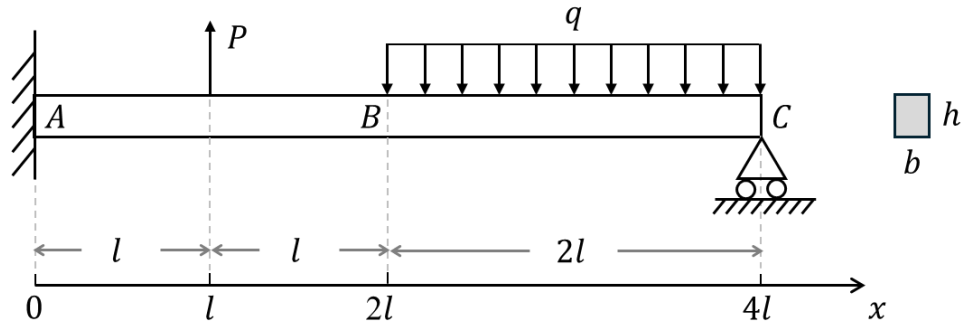
2) check the shear stress

$$\tau_{\max} = 1.5 \frac{F_s}{A} = 1.5 \frac{P/2}{1.5h^2} = \frac{P}{2h^2} \leq [\tau]$$

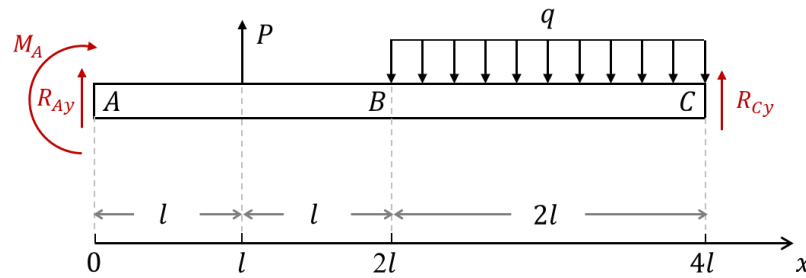
$$h \geq \sqrt{\frac{P}{2[\tau]}} = 63.2\text{ mm}$$

$$\text{So } h = 138\text{ mm} \quad b = 208\text{ mm}$$

2. A straight prismatic beam AD has a rectangular cross-section with width b and height h . Its total length is $4l$, and its Young's modulus is E . The beam is subjected to a concentrated load P and distributed load q per unit length, as shown in the figure below.



- (1) Assuming the transverse reaction force at the left end A is R_{Ay} , the reaction moment at the left end A is M_A , and the transverse reaction force at the right end C is R_{Cy} , as shown below, please write down the force equilibrium equation and moment equilibrium equation.



$$R_{Ay} + R_{Cy} + P - q \cdot 2l = 0 \quad (1a)$$

$$-M_A + P \cdot l - (q \cdot 2l) \cdot 3l + R_{Cy} \cdot 4l = 0 \quad (1b)$$

- (2) Please calculate the internal transverse shear force, $V(x)$, and the internal moment, $M(x)$, at cross-section x for $0 < x < l$.

$$V(x) = R_{Ay} \quad (2a)$$

$$M(x) = M_A + R_{Ay} \cdot x \quad (2b)$$

- (3) Please calculate the internal transverse shear force, $V(x)$, and the internal moment, $M(x)$, at cross-section x for $l < x < 2l$.

$$V(x) = R_{Ay} + P \quad (3a)$$

$$M(x) = M_A - P \cdot l + (R_{Ay} + P) \cdot x \quad (3b)$$

- (4) Please calculate the internal transverse shear force, $V(x)$, and the internal moment, $M(x)$, at cross-section x for $2l < x < 4l$.

$$V(x) = R_{Ay} + P - q \cdot (x - 2l) \quad (4a)$$

$$M(x) = M_A - P \cdot l + (R_{Ay} + P) \cdot x - \frac{q}{2} \cdot (x - 2l)^2 \quad (4b)$$

- (5) What are the two boundary conditions for the deflection curve, $v(x)$, at the left end ($x = 0$)? Please review Table 7.1 in the textbook.

$$v(0) = 0 \quad (5a)$$

$$v'(0) = 0 \quad (5b)$$

- (6) Please calculate the deflection, v , at $x = l$, by combining the moment-curvature equation (Eq. 7.8 in the textbook) and the two boundary conditions.

Substituting Eq. 2b into $EIv'' = M$ gives

$$v''(x) = \frac{M(x)}{EI} = \frac{M_A + R_{Ay} \cdot x}{EI}, 0 < x < l. \quad (6a)$$

where $I = \frac{1}{12}bh^3$.

Integrating Eq. 6a gives

$$v'(x) = \frac{M_A x + \frac{1}{2}R_{Ay} \cdot x^2}{EI} + C_1, 0 < x < l. \quad (6b)$$

$$v(x) = \frac{\frac{1}{2}M_A x^2 + \frac{1}{6}R_{Ay} \cdot x^3}{EI} + C_1 x + C_2, 0 < x < l. \quad (6c)$$

C_1 and C_2 can be determined by substituting Eq. 6b into Eq. 5b and Eq. 6c into Eq. 5a:

$$C_1 = 0, C_2 = 0. \quad (6d)$$

Thus, we have

$$v(x) = \frac{\frac{1}{2}M_A x^2 + \frac{1}{6}R_{Ay} \cdot x^3}{EI}, 0 < x < l. \quad (6e)$$

At $x = l$, we have

$$v(l) = \frac{\frac{1}{2}M_A l^2 + \frac{1}{6}R_{Ay} \cdot l^3}{EI}. \quad (6f)$$

$$v'(l) = \frac{M_A l + \frac{1}{2}R_{Ay} \cdot l^2}{EI}. \quad (6g)$$

- (7) What are the continuity conditions at $x = l$? Please review Table 7.2 in the textbook.

Both v and v' are continuous at $x = l$.

- (8) Please calculate the deflection, v , at $x = 2l$ by combining the moment-curvature equation and the continuity conditions.

Substituting Eq. 3b into $EIv'' = M$ gives

$$v''(x) = \frac{M(x)}{EI} = \frac{M_A - P \cdot l + (R_{Ay} + P) \cdot x}{EI}, l < x < 2l. \quad (7a)$$

Integrating Eq. 7a gives

$$v'(x) = \frac{(M_A - P \cdot l)x + \frac{1}{2}(R_{Ay} + P) \cdot x^2}{EI} + C_3, l < x < 2l. \quad (7b)$$

$$v(x) = \frac{\frac{1}{2}(M_A - P \cdot l)x^2 + \frac{1}{6}(R_{Ay} + P) \cdot x^3}{EI} + C_3x + C_4, l < x < 2l. \quad (7c)$$

C_3 and C_4 can be determined by substituting Eq. 7b into Eq. 6g and Eq. 7c into Eq. 6f:

$$C_3 = \frac{\frac{1}{2}P \cdot l^2}{EI}, C_4 = -\frac{\frac{1}{6}P \cdot l^3}{EI}. \quad (7d)$$

Thus, we have

$$v(x) = \frac{-\frac{1}{6}P \cdot l^3 + \frac{1}{2}(P \cdot l^2)x + \frac{1}{2}(M_A - P \cdot l)x^2 + \frac{1}{6}(R_{Ay} + P) \cdot x^3}{EI}, l < x < 2l. \quad (7e)$$

At $x = 2l$, we have

$$v(2l) = \frac{2M_A l^2 + \left(\frac{4}{3}R_{Ay} + \frac{1}{6}P\right) \cdot l^3}{EI}. \quad (7f)$$

$$v'(2l) = \frac{2M_A l + \left(2R_{Ay} + \frac{1}{2}P\right) \cdot l^2}{EI}. \quad (7g)$$

(9) What are the continuity conditions at $x = 2l$?

Both v and v' are continuous at $x = 2l$.

(10) Please calculate the deflection, v , at $x = 4l$.

Substituting Eq. 4b into $EIv'' = M$ gives

$$v''(x) = \frac{M(x)}{EI} = \frac{M_A - P \cdot l + (R_{Ay} + P) \cdot x - \frac{q}{2} \cdot (x - 2l)^2}{EI}, 2l < x < 4l. \quad (8a)$$

Integrating Eq. 7a gives

$$v'(x) = \frac{(M_A - P \cdot l)x + \frac{1}{2}(R_{Ay} + P) \cdot x^2 - \frac{q}{6} \cdot (x - 2l)^3}{EI} + C_5, 2l < x < 4l. \quad (8b)$$

$$v(x) = \frac{\frac{1}{2}(M_A - P \cdot l)x^2 + \frac{1}{6}(R_{Ay} + P) \cdot x^3 - \frac{q}{24} \cdot (x - 2l)^4}{EI} + C_5x + C_6, 2l < x < 4l. \quad (8c)$$

C_5 and C_6 can be determined by substituting Eq. 8b into Eq. 7g and Eq. 8c into Eq. 7f:

$$C_5 = \frac{\frac{1}{2}P \cdot l^2}{EI}, C_6 = -\frac{\frac{1}{6}P \cdot l^3}{EI}. \quad (8d)$$

Thus, we have

$$v(x) = \frac{-\frac{1}{6}P \cdot l^3 + \frac{1}{2}(P \cdot l^2)x + \frac{1}{2}(M_A - P \cdot l)x^2 + \frac{1}{6}(R_{Ay} + P) \cdot x^3 - \frac{q}{24} \cdot (x - 2l)^4}{EI}. \quad (8e)$$

At $x = 4l$, we have

$$v(4l) = \frac{8M_A l^2 + \left(\frac{32}{3}R_{Ay} + \frac{9}{2}P\right) \cdot l^3 - \frac{2}{3}ql^4}{EI}. \quad (8f)$$

(11) What is the boundary condition for the deflection curve, $v(x)$, at the right end ($x = 4l$)?

$$v(4l) = 0 \quad (9a)$$

Substituting Eq. 8f into Eq. 9a gives

$$\frac{8M_A l^2 + \left(\frac{32}{3}R_{Ay} + \frac{9}{2}P\right) \cdot l^3 - \frac{2}{3}ql^4}{EI} = 0 \quad (9b)$$

(12) Please solve for R_{Ay} , R_{Cy} , and M_A by combining this new boundary condition and the two equilibrium equations obtained in (1). The final answers should be expressed in terms of P, q, l only.

Combining Eq. 9b with Eqs. 1a and 1b gives

$$\begin{cases} R_{Ay} + R_{Cy} + P - q \cdot 2l = 0 \\ -M_A + P \cdot l - (q \cdot 2l) \cdot 3l + R_{Cy} \cdot 4l = 0 \\ 8M_A l^2 + \left(\frac{32}{3}R_{Ay} + \frac{9}{2}P\right) \cdot l^3 - \frac{2}{3}ql^4 = 0 \end{cases} \quad (10a)$$

Reorganizing Eq. 10a gives

$$\begin{cases} R_{Ay} + R_{Cy} = -P + 2ql \\ -M_A + 4lR_{Cy} = -Pl + 6ql^2 \\ 8M_A + \frac{32}{3}lR_{Ay} = -\frac{9}{2}Pl + \frac{2}{3}ql^2 \end{cases} \quad (10b)$$

Solving the linear system of Eq. 10b gives

$$R_{Ay} = -\frac{117}{128}P + \frac{23}{32}ql \quad (10c)$$

$$R_{Cy} = -\frac{11}{128}P + \frac{41}{32}ql \quad (10d)$$

$$M_A = \frac{21}{32}Pl - \frac{7}{8}ql^2 \quad (10e)$$

(13) Please calculate the deflection, v , at $x = 3l$.

Substituting Eqs. 10c, 10d, and 10e into Eq. 8e gives the deflection for $2l < x < 4l$

$$v(x) = \frac{-\frac{1}{6}Pl^3 + \frac{1}{2}Pl^2x - \left(\frac{11}{64}Pl + \frac{7}{16}ql^2\right)x^2 + \left(\frac{11}{768}P + \frac{23}{192}ql\right)x^3 - \frac{q}{24} \cdot (x - 2l)^4}{EI}. \quad (11a)$$

At $x = 3l$,

$$v(3l) = \frac{\frac{133}{768}Pl^3 - \frac{143}{192}ql^4}{EI}. \quad (11b)$$

(14) Please calculate the maximum normal stress on the cross-section at $x = 3l$.

Substituting Eqs. 10c, 10d, and 10e into Eq. 4b gives the internal moment for $2l < x < 4l$

$$M(x) = -\left(\frac{11}{32}Pl + \frac{7}{8}ql^2\right) + \left(\frac{11}{128}P + \frac{23}{32}ql\right) \cdot x - \frac{q}{2} \cdot (x - 2l)^2 \quad (12a)$$

At $x = 3l$,

$$M(3l) = -\left(\frac{11}{32}Pl + \frac{7}{8}ql^2\right) + \left(\frac{11}{128}P + \frac{23}{32}ql\right) \cdot 3l - \frac{q}{2} \cdot l^2 = -\frac{11}{128}Pl + \frac{25}{32}ql^2 \quad (12b)$$

Thus, the maximum normal stress on the cross-section at $x = 3l$ is

$$\sigma_{max}(3L) = \left| \frac{M(3L)}{I} \cdot y_{max} \right| = \frac{|M(3L)|}{\frac{1}{12}bh^3} \cdot \frac{h}{2} = \frac{\left| -\frac{11}{128}Pl + \frac{25}{32}ql^2 \right|}{\frac{1}{6}bh^2} \quad (12c)$$

(15) Please calculate the maximum shear stress on the cross-section at $x = 3l$.

Substituting Eqs. 10c, 10d, and 10e into Eq. 4a gives the internal shear force for $2l < x < 4l$

$$V(x) = \left(\frac{11}{128}P + \frac{23}{32}ql\right) - q \cdot (x - 2l) \quad (13a)$$

At $x = 3l$,

$$V(3l) = \left(\frac{11}{128}P + \frac{23}{32}ql\right) - ql = \frac{11}{128}P - \frac{9}{32}ql \quad (13b)$$

Thus, the maximum shear stress on the cross-section at $x = 3l$ is

$$\tau_{max}(3L) = \frac{3}{2}|\tau_{avg}(3L)| = \frac{3}{2} \left| \frac{V(3l)}{bh} \right| = \frac{3}{2} \frac{\left| \frac{11}{128}P - \frac{9}{32}ql \right|}{bh} \quad (13c)$$